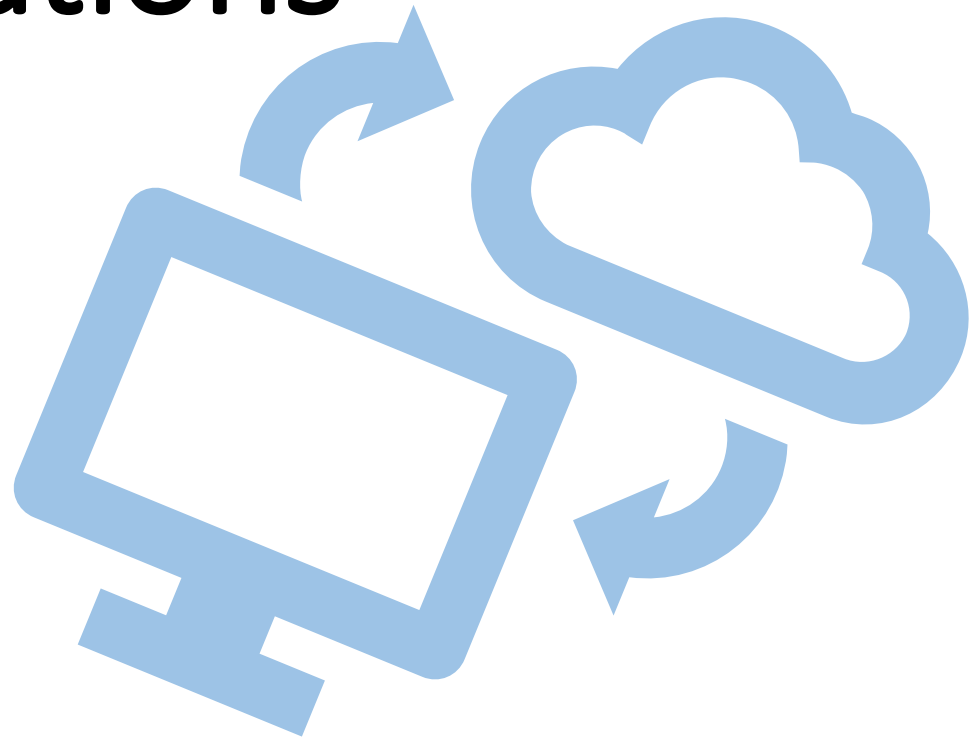
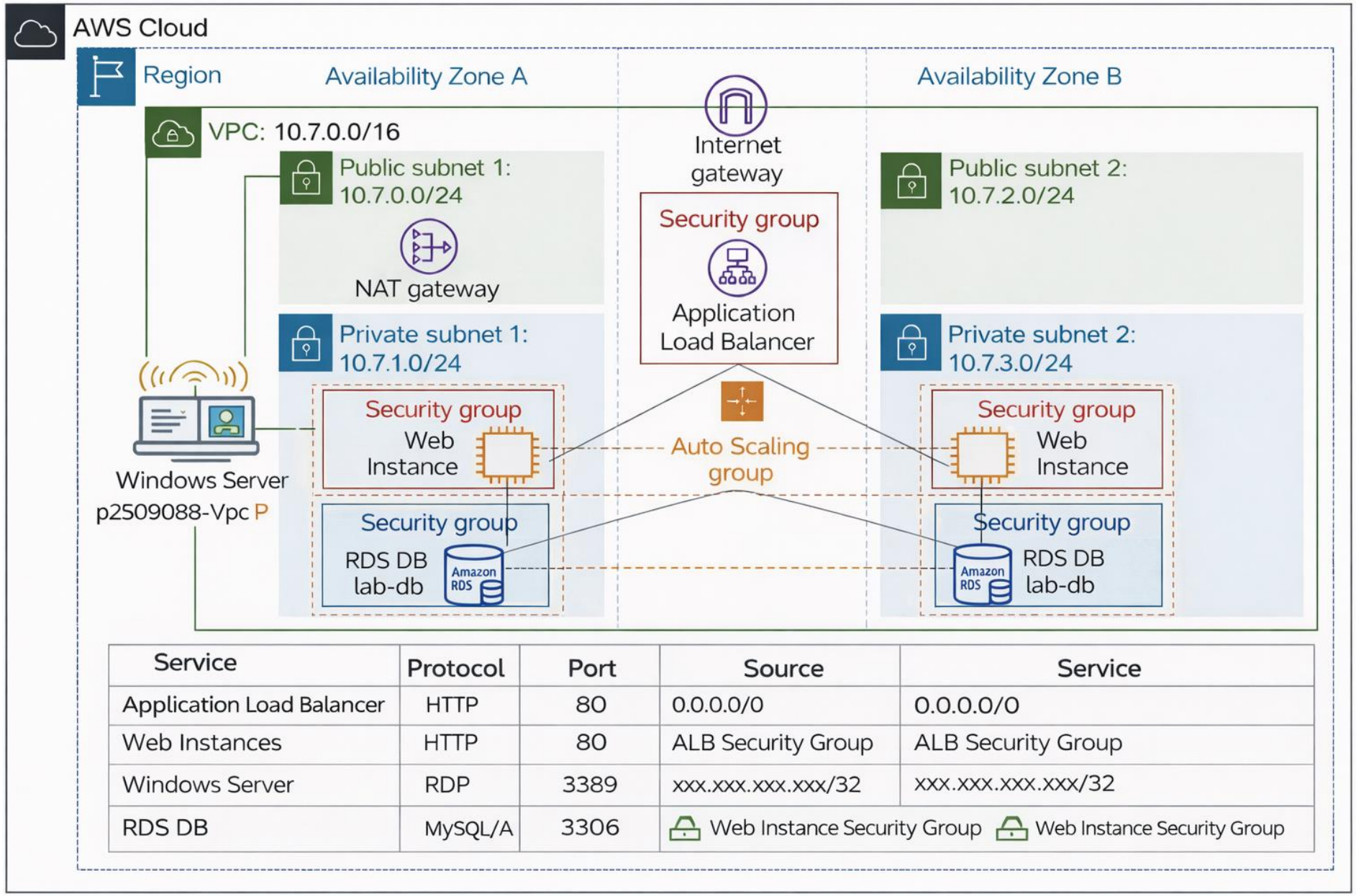
AWS Cloud Foundations Design Document



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Architecture Overview



Service	Protocol	Port	Source	Service
Application Load Balancer	HTTP	80	0.0.0.0/0	0.0.0.0/0
Web Instances	HTTP	80	ALB Security Group	ALB Security Group
Windows Server	RDP	3389	xxx.xxx.xxx.xxx/32	xxx.xxx.xxx.xxx/32
RDS DB	MySQL/A	3306	Web Instance Security Group	Web Instance Security Group

Services Deployed and Justification

Services	Brief Description	Justification
Application Load Balancer (ALB)	Distributes incoming traffic across multiple EC2 instances.	Improves availability and fault tolerance by routing traffic only to healthy targets.
Auto Scaling Group (ASG)	Automatically adjusts the number of EC2 instances based on CPU utilization.	Maintains high availability and ensures CPU utilization does not exceed 70% as required.
Amazon RDS (MySQL)	Managed relational database service deployed in subnets.	Provides reliable, scalable, and managed database solution for application backend.
Detailed CloudWatch Monitoring	Enables 1-minute monitoring metrics for EC2 instances.	Provides real-time performance visibility, allowing rapid scaling decisions and proactive system monitoring in mission-critical government environments.

VPC and Subnet Details

VPC Name:	p2500908-vpc
Region:	United States (N. Virginia)
CIDR Range	10.7.0.0/16

Subnet Name	Type (Public/Private)	AZ	CIDR
p2500908-subnet-public1	Public	AZ-A	10.7.0.0/24
p2500908-subnet-private1	Private	AZ-A	10.7.1.0/24
p2500908-subnet-public2	Public	AZ-B	10.7.2.0/24
p2500908-subnet-private2	Private	AZ-B	10.7.3.0/24

Route Tables Details

Route Table Name	VPC	Descriptions	Subnets	AZ
p2500908-rtb-public	p2500908-vpc	Public routing table	p2500908-subnet-public1 **and** p2500908-subnet-public2	Both AZ-A and AZ-B are included here
p2500908-rtb-private	p2500908-vpc	Private routing table	p2500908-subnet-private1 **and** p2500908-subnet-private2	Both AZ-A and AZ-B are included here

Route Tables Details (Part 2)

Route Table Name:	p2500908-rtb-public
Destination CIDR	Target
0.0.0.0/0	p2500908-igw (Internet Gateway)
10.7.0.0/16	local

Route Table Name:	p2500908-rtb-private
Destination CIDR	Target
0.0.0.0/0	p2500908-natgw (NAT Gateway)
10.7.0.0/16	local

Security Groups (SG) Details

SG Name:	p2500908-sg-alb
Inbound/ Outbound:	Inbound
Type:	HTTP
Source:	Anywhere-IPv4 (0.0.0.0/0)
Description:	Allow public HTTP to ALB (Application Load Balancer), this allows citizen HTTP access

SG Name:	p2500908-sg-windows
Inbound/ Outbound:	Inbound
Type:	RDP
Source:	Anywhere-IPv4 (0.0.0.0/0)
Description:	Allows RDP from admin

Security Groups (SG) Details (Part 2)

SG Name:	p2500908-sg-web
Inbound/ Outbound:	Inbound
Type:	SSH
Source:	My IP
Description:	Allows Admin SSH

SG Name:	p2500908-sg-web
Inbound/ Outbound:	Inbound
Type:	HTTP
Source:	Custom (p2500908-sg-alb)
Description:	Allows ALB to web tier only

Security Groups (SG) Details (Part 3)

SG Name:	p2500908-sg-rds
Inbound/ Outbound:	Inbound
Type:	MySQL/Aurora
Source:	Custom (p2500908-sg-web)
Description:	Allow MySQL from web tier

SG Name:	p2500908-sg-web
Inbound/ Outbound:	Inbound
Type:	HTTP
Source:	Custom (10.7.0.0/16)
Description:	Allow internal VPC traffic

Note: All of the Outbound Rules are left as default.

EC2 Instance Details

Tag	OS	Type	Size	Public IPv4 address	Private IPv4 address	Which Subnet	User Data (Y/N)*
p2500908-HelloWebPage	Amazon Linux 2023 kernel-6.1	t3.micro	8 GiB gp3 Root volume, 3000 IOPS, Not encrypted	Nil	10.7.1.81	p2500908-subnet-private1	Y
p2500908-WindowsServer	Windows_Server-2016-English-Full-Base-2026.02.11	t3.micro	30 GiB gp3 Root volume, 3000 IOPS, Not encrypted	54.221.85.81 (Associated Elastic IP)	10.7.0.218	p2500908-subnet-prublic1	N

User Data for EC2 instances (p2500908-HelloWebPage)

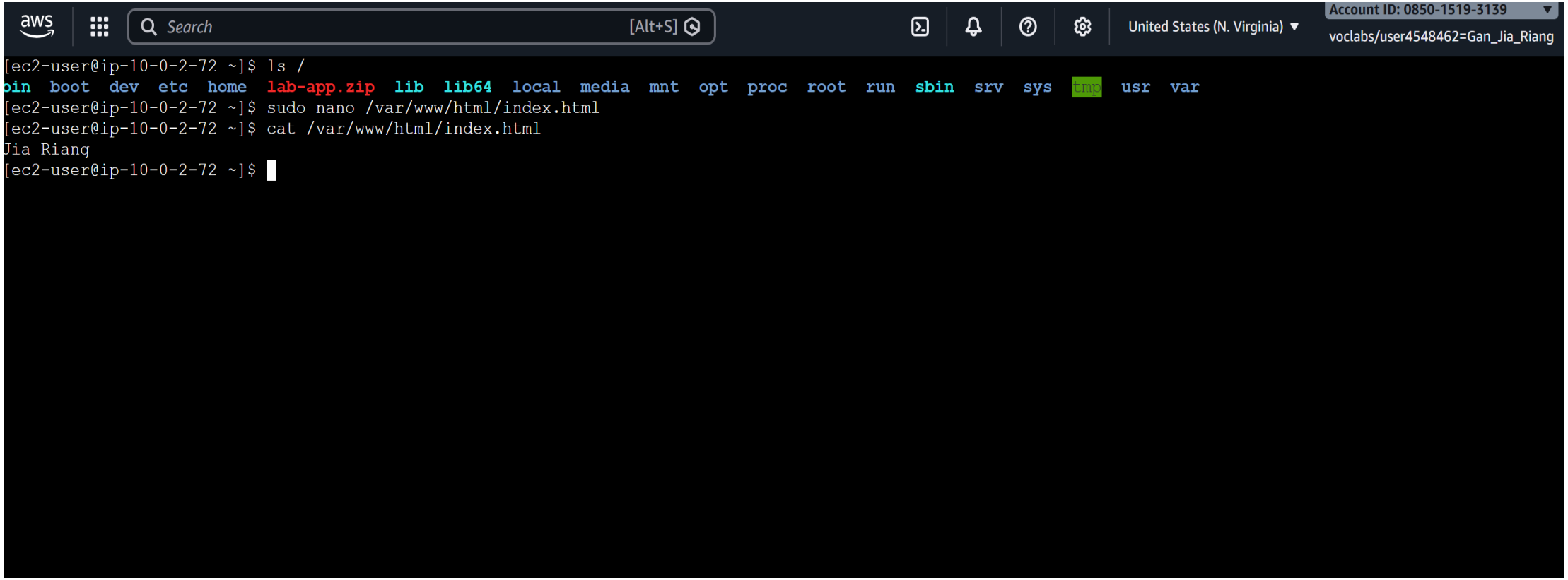
User data – *optional* | Info

Upload a file with your user data or enter it in the field.

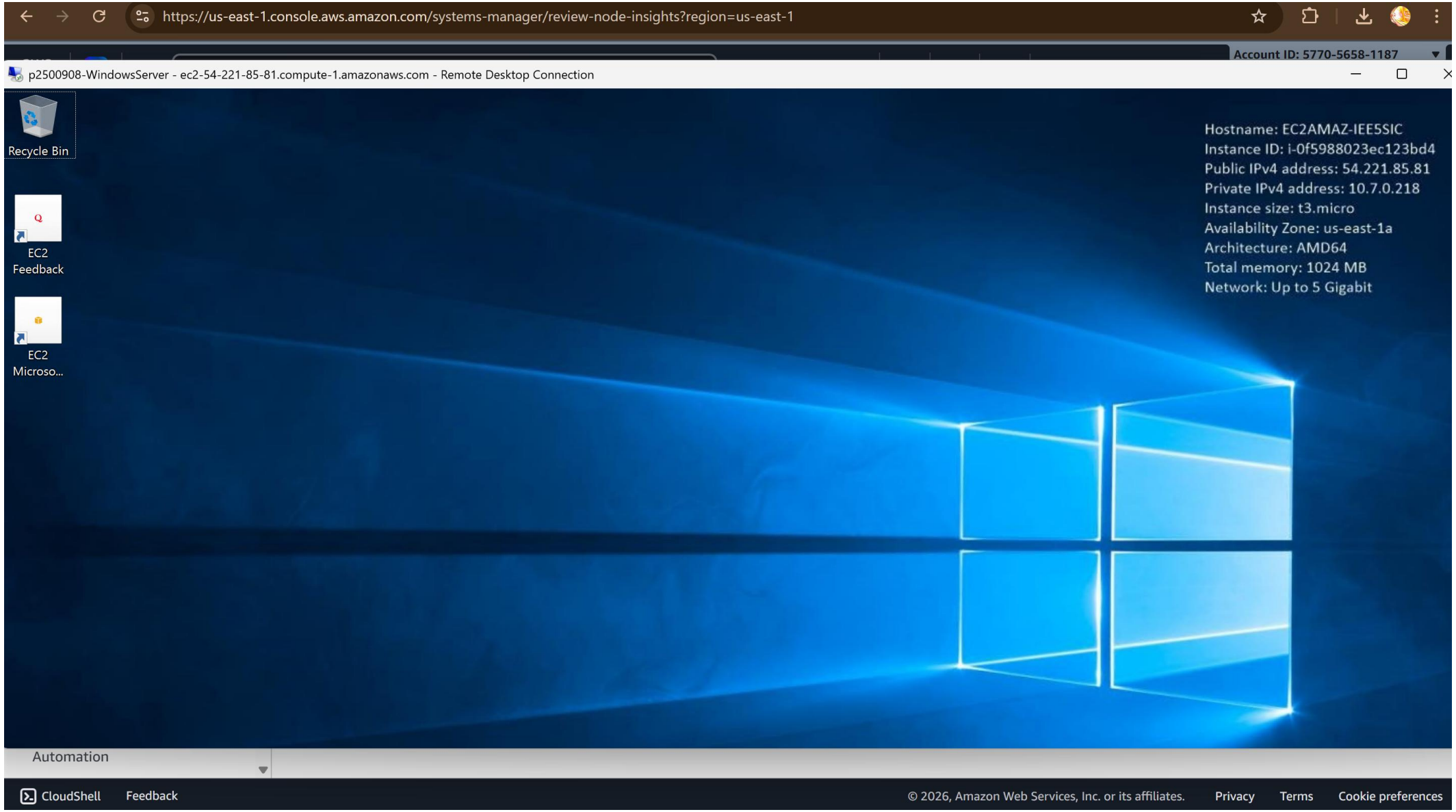
Choose file

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' >
/var/www/html/index.html
```

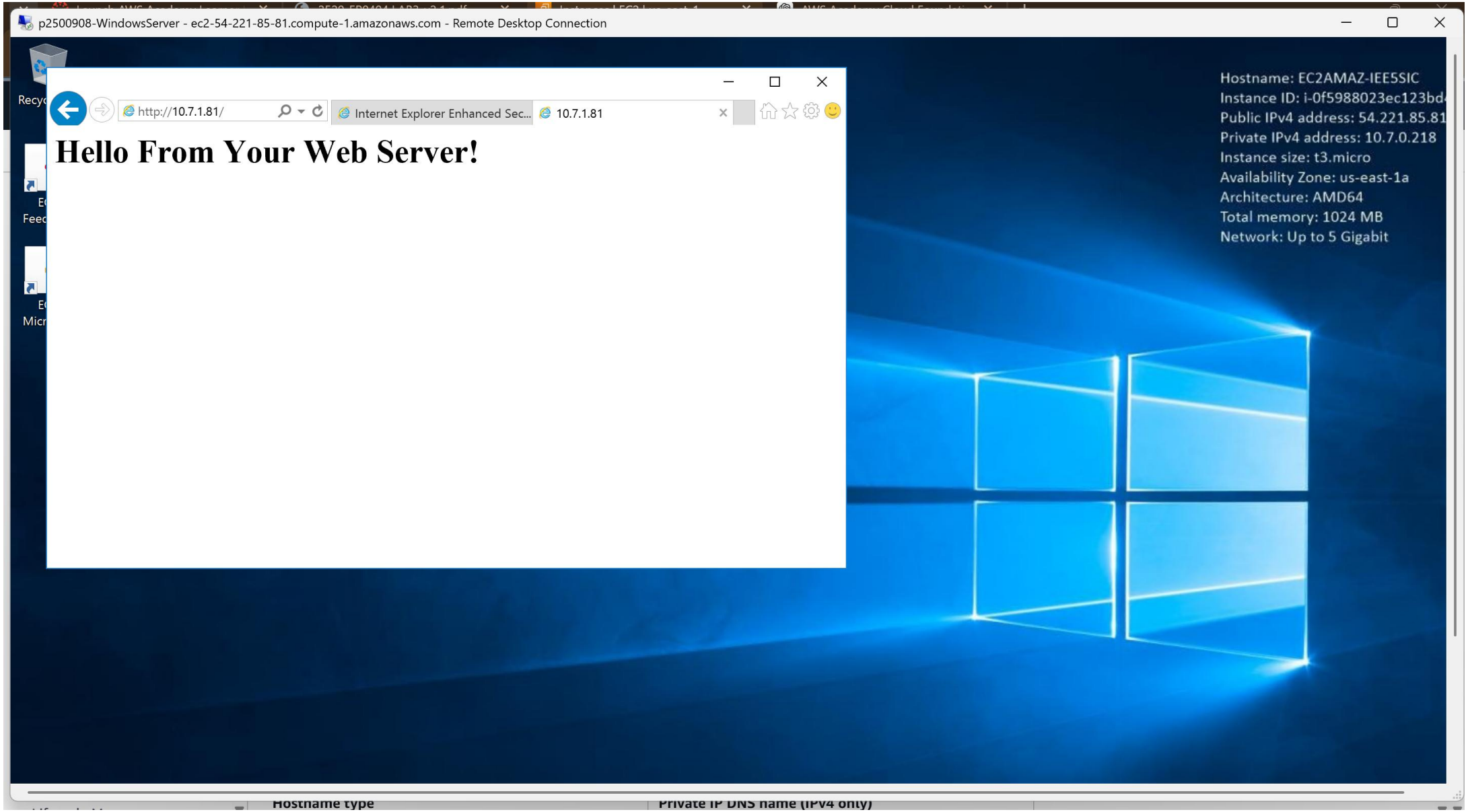
SSH Connection details



RDP Connection details



RDP Connection details (Part 2)



RDS details

- **Name:** lab-db
- **Subnets:** p2500908-subnet-public1, p2500908-subnet-public2

- **Engine:** MySQL
- **Type:** Dev/Test
- **DB Instance Class:** db.t3.micro

- **VPC:** p2500908-vpc
- **Public access:** YES
- **DB Subnet Group:** p2500908-dbsubnetgrp
- **VPC Security Group:** p2500908-sg-rds

Load Balancer Details

Name	Tag	Type (External/ Internal)	SG Name
p2500908-alb	**Key:** Application Load Balancer **Value:** P2500908	Application	p2500908-sg-alb

Auto Scaling Launch Configuration Details

Name	OS	Type	Size	Role	SG Name
p2500908-lt-web	Amazon Linux 2023	t3.micro	NA	Web Server (Apache)	p2500908-sg-web

Auto Scaling Group Details

Name	Launch Configuration	Group Size	VPC	Subnets	ELB
p2500908-asg-web	p2500908-lt-web	**Desired capacity:** 2 **Minimum capacity:** 2 **Maximum capacity:** 4	p2500908-vpc	p2500908-subnet-private1, p2500908-subnet-private2	p2500908-alb (Target Group: p2500908-tg-web)